

# UGESH R

Data Engineer · Azure & Databricks · PySpark / SQL / Airflow · Generative AI

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## TECHNICAL SKILLS

**Languages:** Python, SQL, Spark SQL

**Cloud & Platforms:** Azure Databricks, Azure Data Factory (ADF), ADLS Gen2, Azure SQL Database, Microsoft Azure

**Data Engineering:** Apache Spark, PySpark, Delta Lake, Apache Airflow, Hadoop / Cloudera (CDP), Medallion Architecture, CI/CD

**Generative AI:** Prompt Engineering, Retrieval-Augmented Generation (RAG), LangChain, Azure OpenAI Service, Vector Embeddings & Vector Search (FAISS)

**Tools:** Git, Power BI, JIRA

## EXPERIENCE

### Cognizant Technology Solutions

Jul 2022 – Present

Data Engineer – Associate

Tamil Nadu, India

- Migrated 10+ Informatica PowerCenter mappings to Azure Databricks using PySpark and Spark SQL, improving pipeline performance and eliminating on-premise infrastructure dependency entirely.
- Re-engineered 25+ SSIS-based ETL workflows into Azure Databricks pipelines using PySpark, ADF, and ADLS Gen2, enabling cloud-native scalability and automated incremental loads.
- Architected and maintained Azure Data Factory orchestration across all migrated workloads, including scheduling and failure/error handling to reduce manual monitoring effort.
- Engineered PySpark transformation jobs on Databricks backed by Delta Lake to process structured and semi-structured data at scale.
- Modeled Bronze/Silver/Gold Lakehouse layers in ADLS Gen2; enforced schema validation and business rules, strengthening data quality and downstream trust in reporting.
- Delivered analytics-ready datasets and Power BI reports consumed directly by data science and business analyst teams for decision-making.

### The Reciprocal Solutions

2021 – 2022

Junior Developer – Intern

Tamil Nadu, India

- Developed internal web application modules and produced wireframes/site maps supporting a full website revamp.

## PROJECTS

### GenAI-Powered Lakehouse Q&A Assistant GenAI + PySpark · Databricks · Delta Lake · LangChain · Azure OpenAI ·

DE

FAISS

- Built an end-to-end pipeline: ingested raw retail sales data into ADLS Gen2, transformed it through PySpark on Azure Databricks into Bronze/Silver/Gold Delta tables.
- Embedded Gold-layer schema and business metadata into a FAISS vector store, and built a LangChain + Azure OpenAI RAG assistant that answers natural-language questions (e.g. "top products last quarter") by generating and executing Spark SQL against the Lakehouse.
- Enabled self-serve, natural-language analytics for non-technical users, reducing dependency on manual ad-hoc reporting requests.

### ADF + ADLS Gen2 End-to-End Pipeline

ADF · ADLS Gen2 · Azure SQL

- Designed a complete ingest-transform-serve pipeline: raw data landed in ADLS Gen2 via ADF, transformed in Azure SQL, and served to a reporting layer.

### PySpark Transformation Pipeline on Databricks

PySpark · Databricks · Delta Lake · Spark SQL

- Built a batch pipeline to ingest, clean, and transform large-scale retail sales data on Azure Databricks, persisted in Delta Lake for downstream analytics.

## EDUCATION

## **I CERTIFICATIONS**

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Microsoft Certified: Azure Data Engineer Associate (DP-203) – *In Progress* | PySpark Fundamentals (Udemy) |  
Databricks Basics for Data Engineering (Udemy)